

ZAHRA KHATTI

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About Me

Ph.D. student in Industrial and Systems Engineering specializing in large-scale stochastic optimization. Experienced in designing algorithms for adaptive sampling and fairness-aware machine learning, with strong skills in mathematical analysis, neural network implementation, and optimization under constraints. Proficient in Python and PyTorch, with research interests spanning optimization algorithms, modeling, and applied machine learning.

Skills

- **Mathematical:** Optimization, Operations Research, Machine Learning, Probability, Statistics
- **Computational:** Python, PyTorch, Gurobi, AMPL, R, Git, Arena, GAMS

Experience

Research Assistant

Aug. 2023 – Present

Lehigh University, Department of Industrial and Systems Engineering

Bethlehem, PA

- Conducting research on **adaptive sampling methods** for stochastic optimization, focusing on dynamically adjusting sample sizes across iterations to improve efficiency and convergence guarantees.
- Developing theoretical proofs on algorithm convergence under various assumptions and analyzing bounds.
- Designed a new fairness measure and applied it to large-scale machine learning problems with nonconvex constraint-based formulations.
- Implemented neural networks in **PyTorch** and applied **SQP-based optimization** for constrained machine learning.

Marketing Strategist

Jan. 2018 – Jul. 2019

K. N. Toosi Innovation Center

Tehran, Iran

- Led marketing strategies for a sports-focused startup, increasing user engagement and brand awareness.
- Analyzed market trends and customer feedback to guide strategic decision-making.

Education

Lehigh University

Aug. 2023 – Present

PhD in Industrial and Systems Engineering

Bethlehem, PA

- Advisors: Prof. Frank E. Curtis, Prof. Daniel P. Robinson
 - Research Focus: Large-Scale Optimization Algorithms for Nonlinear Problems with Applications in Machine Learning and Data Science.

Sharif University of Technology

2019 – 2022

Master of Science in Industrial Engineering, Economic and Social Systems

Tehran, Iran

- Advisor: Prof. Farhad Kianfar
 - Dissertation: Customer Segmentation and Migration Pattern Modeling Based on Loyalty.

K.N. Toosi University of Technology

2015 – 2019

Bachelor of Science in Industrial Engineering

Tehran, Iran

Publications

1. **Zahra Khatti, Daniel P. Robinson, and Frank E. Curtis.** Fair supervised learning through constraints on smooth nonconvex unfairness-measure surrogates. *arXiv preprint arXiv:2505.15788*, 2025.

Leadership and Service

MOPTA Conference Co-chair

2026 (ongoing)

Lehigh University

Bethlehem, PA

- Co-organizing the 2026 MOPTA conference; coordinating speakers, sessions, and logistics.

GradOpt and ColOpt workshops Co-chair

2026 (ongoing)

Lehigh University

Bethlehem, PA

- Organizing the events related to inviting master or undergrad students interested in this field

Graduation Ceremony Organizer

2019

K.N. Toosi University of Technology

Tehran, Iran

Executive Board Member

2016–2017

Scientific Association, K.N. Toosi University

Tehran, Iran

Selected Courses

Machine Learning · Neural Networks, Convolutional Neural Networks, Optimization Algorithms (SGD, Adam), Convergence Theorems
Statistical Machine Learning · Supervised Learning, Linear & Logistic Regression, Decision Trees, SVM, Ensemble Methods (Random Forest, AdaBoost, XGBoost)
Continuous Optimization · Gradient-based Methods, Constrained Optimization, Nonlinear Programming, Lagrangian Methods
Project Management · Planning, Scheduling, Risk Management, Resource Allocation
Decision Management · Bayesian Decision Analysis, Multi-criteria Decision Analysis

Honors and Awards

- Full Scholarship from Sharif University (M.Sc.)
- Selected as an “Exceptionally Talented” Student, Department of Industrial Engineering, Sharif University
- Ranked 3rd among 42 students in B.Sc., Department of Industrial Engineering, K.N. Toosi University
- Member of the Scientific Association, Winner of the 10th “Movement” Festival
- Full Scholarship from K.N. Toosi University (B.Sc.)

Selected Projects

Review of Uncertainty Quantification for Deep Learning Models

2023

Lehigh University

Bethlehem, PA

- Reviewed and analyzed a research paper on uncertainty quantification techniques for deep learning regression models under limited data conditions.

Business Process Simulation Models from Event Logs

2020

Sharif University of Technology

Tehran, Iran

- Developed a model for automated discovery of business process simulation models based on event log data.

Multi-Stage Financial Planning Problem

2020

Sharif University of Technology

Tehran, Iran

- Applied scenario reduction and nested decomposition methods to optimize a multi-stage financial planning problem.